InsightStream: Cloud Data Warehouse & Analytics

Building a Modern ELT Pipeline with Airflow, Snowflake Internal Storage, dbt, and Power BI

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Project Overview

Project Proposal

Overview of the project, objectives, and scope for building a robust, modern ELT pipeline for data analytics using Snowflake's native capabilities and industry-standard transformation tools, focusing on large-scale **Dataset Ingestion** and complex data modeling.

Business Objectives

- Establish a comprehensive ELT pipeline for ingesting diverse **Datasets** directly into a cloud data warehouse.
- Enable advanced analytics and business intelligence through **Snowflake** and **Power BI** for data-driven decision making.
- Automate data workflows using **Airflow** and **dbt** to ensure data freshness, reliability, and scalability.

Technical Objectives

- **Data Ingestion:** Ingest large-scale **Datasets (CSV, Parquet)** directly into **Snowflake Internal Storage** (Internal Stages) for raw data storage.
- **Data Transformation & Warehousing:** Utilize **dbt (data build tool)** to perform complex data cleaning, modeling, and transformation within Snowflake, ensuring modularity and automated documentation.
- **Workflow Orchestration:** Implement **Airflow DAGs** to manage and automate the end-to-end ELT process.
- **Business Intelligence:** Connect Snowflake to **Power BI** for creating powerful, interactive dashboards and enterprise-grade visualizations.
- **Monitoring & Automation:** Set up monitoring for Airflow DAGs and dbt runs, ensuring data quality and pipeline health.
- **Documentation:** Provide comprehensive documentation for the architecture, dbt models, and Power BI reports.

Project Documentation Guidelines

Item	Student Deadline	Graduate Deadline
Project Planning & Management	2/20/2026	2/3/2026
Literature Review	4/20/2026	3/1/2026
Requirements Gathering	4/20/2026	3/1/2026
System Analysis & Design	5/1/2026	4/3/2026
Implementation (Source Code & Execution)	7/10/2026	5/17/2026
Final Presentation & Testing & Reports	7/17/2026	5/24/2026

LITERATURE REVIEW

Researching Best Practices for Modern ELT Pipelines

- Techniques for ingesting large-scale **Datasets (e.g., CSV, Parquet)** directly into **Snowflake Internal Stages**, optimizing for performance and cost.
- Best practices for data modeling using **dbt**, focusing on Modular SQL, Macros, and automated testing frameworks.
- Effective use of **Apache Airflow** for orchestrating dbt jobs and data ingestion tasks.
- Designing compelling and interactive dashboards using **Power BI** for high-impact business intelligence.
- Strategies for ensuring **Data Quality and Governance** across the pipeline using dbt tests and Snowflake features.

Requirements Gathering Details

Functional

- Integration of diverse **Dataset sources**.
- Extraction of specific data points for business logic.
- Complex transformations using **dbt models**.
- Interactive analytical outputs in **Power BI**.

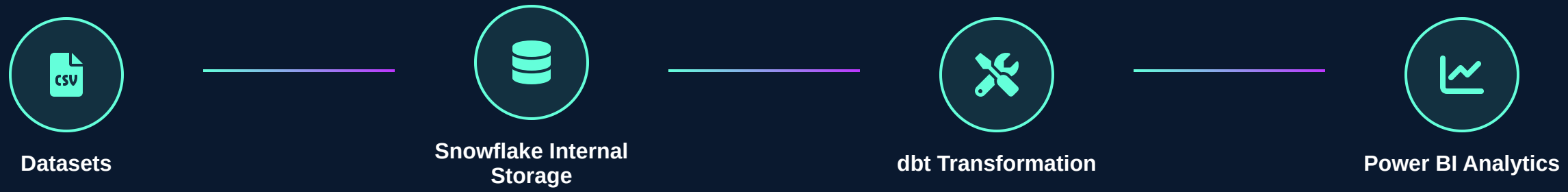
Non-Functional

- Low **data latency** and high throughput.
- Robust **data security** within Snowflake.
- Cost optimization for cloud resources.
- System scalability and disaster recovery.

Data

- Identification of all **Dataset formats**.
- Support for multiple data formats (JSON, CSV, Parquet).
- Handling varying **data volumes**.
- Defined frequency of updates for all sources.

System Analysis & Design



Orchestrated by Apache Airflow

System Architecture

A modern ELT approach where data is ingested directly into **Snowflake Internal Stages** from various **Datasets**. This eliminates external dependencies and reduces latency for raw data availability.

Data Model (dbt)

Utilizing **dbt** to manage the transformation layers: **Staging** (cleaning), **Intermediate** (joining), and **Marts** (business logic), all optimized for analytical performance.

Implementation: Tools & Technologies

Python

Core language for **Dataset processing** and Airflow DAG development.

Snowflake

Cloud Data Warehouse utilizing **Internal Storage** for efficient raw data staging.

dbt

Data Build Tool for SQL-based transformations, version control, and automated documentation.

Apache Airflow

Workflow orchestration for scheduling and monitoring the end-to-end ELT pipeline.

Power BI

Enterprise BI tool for creating interactive dashboards and data visualizations.

Local/Cloud Datasets

Diverse data sources (e.g., CSV, Parquet) integrated into the pipeline for ingestion.

Implementation Plan

01 Cloud Environment Setup

Configure Snowflake accounts, internal stages, and the Apache Airflow environment.

02 Data Ingestion to Snowflake

Develop Python scripts to load **Datasets** and upload directly to Snowflake Internal Stages.

03 dbt Project Setup

Initialize dbt, configure profiles, and define sources.

04 Transformation Layer

Write dbt models to transform raw data into analytical tables.

05 Airflow DAG Development

Design DAGs to orchestrate ingestion scripts, dbt runs and tests.

06 Power BI Dashboard Creation

Develop interactive dashboards connected to Snowflake.

Final Presentation, Testing & Reports

Testing Procedures

Unit Testing

Test individual Python scripts and **dbt models** for logic accuracy.

Data Quality Testing

Use **dbt tests** (unique, not_null, relationships) to ensure data integrity.

Integration Testing

Verify end-to-end data flow from source **Datasets** through Snowflake/dbt into Power BI.

Performance Testing

Evaluate **dbt run times** and Power BI query performance for scalability.

Presentation & Reports

Final Presentation

Showcase project architecture, **dbt lineage**, and key analytical insights.

Technical Report

Comprehensive documentation including **auto-generated dbt docs** and Power BI designs.

Live Demonstration

A live walkthrough showing data flowing from source to interactive dashboards.

Presentation & Reports Details

Final Presentation

- Showcase the project's **modern ELT architecture**.
- Visualize **dbt lineage graphs** for data transparency.
- Present key analytical insights derived from **Power BI**.
- Discuss challenges overcome during implementation.

Technical Report

- Comprehensive documentation of **design choices**.
- Inclusion of **auto-generated dbt documentation**.
- Detailed **Power BI dashboard designs** and logic.
- Cloud service configurations and security protocols.

Live Demonstration: End-to-End Data Flow

This project aims to deliver a robust, scalable, and modern ELT solution for data analytics, leveraging the power of **Snowflake**, **dbt**, **Airflow**, and **Power BI** to provide data-driven insights and support informed decision-making, with a focus on **large-scale Dataset processing**.

THANK YOU!

InsightStream Project Team